Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/02/2011
PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	Risk Assessment of Space Radiation-Enhanced Colon Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	10/01/2008	**End Date:**	09/30/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: NCE to 9/30/2012 per C. Guidry/JSC and NSSC information (Ed., 9/27/2011)		
Key Personnel Changes/Previous PI:	Dr. Story was involved in this project for the first three years. As part of a NCE, his involvement will no longer be required.		
COI Name (Institution):	Roig, Andres (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX08BA54G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal addresses the Carcinogenesis category of the Bioastronauts Roadmap by investigating carcinogenic risk of human colonic cells and tissues after low and high LET radiation exposure. The colon is a major target for the carcinogenic effects of radiation. Aberrant crypt foci are an early precursor lesion for colorectal cancer (CRC) that is characterized by trisomy 7 (present in 37% of adenomas and 80% of adenocarcinomas). The appearance of the adenomatous polyp becomes more prevalent with increased age (10% incidence at age 40 and >25% by age 60). Pre-cancerous lesions may already exist or develop in astronauts going on long-term missions, the result of which would be an increased risk for cancer development through exposure to space radiation. Our key objective is to determine the probabilities for increased colonic carcinogenesis from space radiation as a result of radiation dose, quality, and dose rate. We will expose 2D and 3D human colonic cells and human colonic biopsies to energetic protons (H), high Z and energy particles (HZE), and x-rays and analyze results using both biological assays and genome-wide systems approaches. We are comparing these results to a novel immortalized human trisomy 7 colonic epithelial cell line and will model carcinogenic risk of these cells compared to normal and immortalized diploid colonic cells. We have also initiated a small molecule screen to identify non toxic compounds that may inhibit the trisomy 7 cells but not the normal diploid. We has also examined DNA damage sensing, signaling, and repair kinetics in normal and trisomy 7 human colonic cell lines. Finally, we have obtained a substantial amount of information about a novel oral bioavailable compound CDDO-Me (bardoxolone methyl, RTA 402) and have determined it is an effective and non toxic countermeasure for protecting normal colonic cells from the detrimental effects of low and high LET irradiation. As part of a new project we are extending these countermeasure studies to mouse models. This research should improve our mechanistic understanding for the variability of radiation dose, quality, and dose-rate dependencies for colon cancer development and provide biological markers for modeling and risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of methods to maintain human colonic epithelial cells in culture long term has many benefits for life on Earth. The unique human colonocyte reagents we have developed may help in our understanding of the earliest events in the development of human colon cancer. In addition to normal human colonocytes, we have also isolated the first pre-cancerous human colonocyte cell culture that has one extra chromosome (trisomy 7). Trisomy 7 occurs in a large fraction of patients with sporadic colon cancer and is believed to be one of the earliest steps in colon cancer progression. Our goal is to identify the underlying molecular mechanisms leading to early genetic changes in colon cancer and to validate novel genetic alterations that may occur as critical steps in colon cancer progression. We have initiated experiments to silence and express candidate genes known to be altered in colon cancer into our colonocyte cell lines. Experiments conducting genome-wide sequencing of colon tumors revealed that in addition to the common already known gene mutations, many additional genes are mutated in colon cancer. It is not known which of these mutated genes are important for colon cancer progression. Development of an increasingly cancerous phenotype in the colonocytes after specific genetic alterations should help validate the currently known pathways leading to colon cancer and identify new targets for therapeutic interventions. This work has mostly been accomplished and additional studies are required to complete them. We are also investigating a novel biological countermeasure that may protect against radiation damage. This would also have a major impact on cancer patients being treated with therapeutic radiation.
	1. DNA damage in colonic epithelial cells is more difficult to repair when caused by iron particles compared to protons and gamma rays. (published, see lists at end of this report) 2. Iron damage persists longer in both 2D and 3D colonic epithelial cell cultures compared to protons and gamma rays. (published, see lists at the end of this report) 3. Epithelial cells in intact human colonic tissues display more damage foci than surrounding stromal cells after both high and low LET radiations. All cells in intact colonic tissues are exposed to the same radiation dosage and are likely experiencing equal amounts of double strand breaks, but location in the cell cycle may explain why the replicating epithelial cells show more pronounced DNA damage response compared to quiescent stromal cells. (published, see lists at the end of this report) 4. Additional cells from another patient have been established and characterized. We have an immortalized normal diploid cell line and one that is immortalized with one chromosomal change (trisomy 7). Trisomy 7 is present in about 65% of all colonic adenoma and adenocarcinomas. We have now progressed these cells partially toward tumorigenicity by reducing p53 expression (shRNA to p53), overexpressing K-ras and c- myc and are currently testing if irradiation converts these cells to cancer forming cells. (presented at the St Goar, Satellite meeting, July 2009, manuscript published, see lists at the end of this report). 5. We have now demonstrated and published (Gastroenterology, 138:1012-1021, 2010) the ability to differentiate the immortalized human colonic epithelial cell lines We showed that our human colonic epithelial cells express several gastrointestinal putative stem cell markers including CD44, CD29, BMI-1, and LGR5. This indicates that our immortalized human colonic epithelial cells can make similar 3D differentiated structures as do primary cells from mouse small intestine. Our cells are clonal, can divide long-term (are immortalized), and single cells divide to generate self-organizing structures with several differentiated cell types and architecture of crypts present in adults in the absence of a cellular niche. Our cells demonstrate polarized villin staining, junction formation (ZO-1), a subset of cells express Muc2 similarly to the staining of mouse small intestinal cells, and some even express chromogranin A (a neuroendocrine marker). This also helps demonstrate that we are culturing human colonic epithelial stem cells. 6. We have also passaged the cells with trisomy 7 (contain hTERT, cdk4, shRNAp53 and mutant K-ras) and observed that a subclone of these cells contain a trisomy 20 (also very common in colon cancer). We have also observed a non random trisomy 20 in a HCECs + 7 without addition of mutant K-ras and shRNA to p53. While these cells make large colonies and grow anchorage independent in soft agar they do not make tumors in nude mice. We have now expressed c-myc in these cells and these also do not make tumors. These isogenic partially progressed cell lines are extremely useful for radiation experiments to determine biological steps in cancer progression and new experiments are in progress. Publications and Meeting Abstracts: Articles in Peer-reviewed Journals 1. Roig, AI, Hight, SK, and Shay, JW. "Two and three dimensional models for risk assessment of radiation-enhanced colorectal tumorigenesis" Radiation Research, 171: 33-40, 2009 PMID: 19138051 2. Roig, AI, Wright, WE, Shay, JW. "Is telomerase a novel target for metastatic colon cancer." Current Colorectal Cancer Reports, 5: 203-208, 2009.

Task Progress:	3. Roig, AI, Hight, SK, Minna, JD, Shay, JW, Rusek, A, Story, M. “DNA damage intensity in fibroblasts in a 3 dimensional collagen matrix correlates with the Bragg curve energy distribution of a high LET particle”. International J. Radiation Biology, 86: 194-204, 2010. PMID: 20201648. 4. Roig, A.I., Hight, S.K., Eskiocak, U., Delgado, O., Souza, R, Spechler, S, Wright, W.E. and Shay, J.W. "Experimentally immortalized epithelial cells derived from non cancerous human colon biopsies exhibit stem cell markers". Gastroenterology, 138:1012-1021, 2010. PMID: 19962984 5. Eskiocak, U., Kim, S.B, Roig, A.I., Kitten, E., Batten, K., Cornelius, C., Zou, Y.S., Wright, W.E. and Shay, J.W. "CDDO-Me protects against space radiation-induced transformation of human colonic epithelial cells." Radiation Research, 174: 27-36, 2010. PMID: 20681796 6. Roig, A. and Shay, J.W. "Immortalization of adult human colonic epithelial cells extracted from normal tissues obtained via colonoscopy." Nature Protocols Network (DOI: 10.1038/nprot. 2010.29) 7. Ly, P., Eskiocak, U., Kim, S.B., Roig, A.I., Hight, S.K.,Lulla, D, Zou, Y.S., Batten, K., Wright, W.E., and Shay, J.W. “Characterization of emerged aneuploid populations with non-random trisomy 7 and 20 derived from diploid human colonic epithelial cells”. Neoplasia, 13:348-357, 2011. PMID: 21472139 8. Eskiocak, U., Kim, S.B., Ly, P., Roig, A.I., Biglione, S., Komurov, K., Cornelius, Wright, W.E, White, M.A., and Shay, J.W. “Functional parsing of driver mutations in the colorectal cancer genome reveals numerous suppressors of anchorage-independent growth.” Cancer Res. Priority Reports, in press, 2011 PMID: 21527559 Meeting Abstract and Presentations 1. Eskiocak U, Roig, AI, Kitten, E., Lulla, D, Wright, WE, Shay, JW. "RTA402 protects against radiation induced neoplastic transformation of normal human colonic epithelial cells." Presented at the NASA Human Research Program Investigator's Workship, League City, Texas, February 2-4, 2009. Meeting abstract, platform presentation, Feb-2009 2. Eskiocak U, Roig, AI, Kitten, E., Lulla, D, Wright, WE, Shay, JW. "RTA402 protects against radiation-induced neoplastic transformation of normal human colonic epithelial cells." Presented at the Heavy Ions in Therapy and Space Symposium, 5th International Workshop on Space Radiation Research, Cologne, Germany, July 7-10, 2009. Meeting abstract, platform presentation, Jul-2009 3. Roig, A, Eskiocak U, Hight, S, Lulla, D, Zou, Y, Wright, WE, Shay, JW. “Immortalization and cytogenetic characterization of human colonic epithelial cells” Presented at the 9th International Symposium on Chromosomal Aberrations (ISCA), July 11-12, 2009 Burg Rheinfels/St Goar, Germany. Meeting abstract, platform presentation, Jul-2009 4. Eskiocak U., Kim S.B., Roig A.I., Kitten E., Batten K., Cornelius C., Zou Y.S., Wright W.E., and Shay J.W. " CDDO-Me Protects Against Heavy Ion-Induced Transformation of Human Colonic Epithelial Cells." NASA Human Research Program, Houston, February 3-5, 2010 (JW Shay, session chair). 5. Kim, SB, Eskiocak, U., Roig,A.I., Delgado, O., Kitten, E., Batten, K, Wright, W.E. and Shay, J.W. Bardoxolone methyl (BARD) is both a countermeasure and mitigator of radiation-induced neoplastic transformation of normal human epithelial cells. 21st Annual NASA Space Radiation Investigator's Workshop, May 17-19, 2010. (JW Shay meeting organizing committee, participated in red-blue team debates on whether NASA investigators are making progress in achieving NASA priority gaps, organized poster/oral session contest for graduate students and postdoctoral trainees). 6. Kim, S.B., Eskiocak, U., Cornelius, C., Ly, P., Biglione, S., and Shay, J.W. Bardoxylone methyl: a novel radiation countermeasure and mitigator. 18th IAA meeting in Houston, JW Shay, oral presentation, 2011. 7. The Committee on the Evaluation of Space Radiation Cancer Risk Models, The Keck Center for the National Academies, Overview of the the Lung and GI Tract NASA Research. June 13-15, 2011. JW Shay, oral presentation.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Kim SB, Eskiocak U, Cornelius C, Ly P, Biglione S, Shay JW. "Bardoxolone methyl: a novel radiation countermeasure and mitigator." Presented at 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Eskiocak U, Kim SB, Ly P, Biglione S, Komurov K, Cornelius C, Wright WE, White MA, Shay JW, Roig AI. "Stable Down-Regulation of Putative Colorectal Cancer Candidate Genes Identifies Extensive Fragility in the Matrix-Dependent Proliferation Control Network." Poster presented at DDW (Digestive Disease Week) 2011 Meeting, Chicago, IL, May 7-10, 2011. DDW (Digestive Disease Week) 2011 Meeting, Chicago, IL, May 7-10, 2011. (poster received distinction status). http://ddw.scientificposters.com/epsabstract.cfm?id=1 , May-2011
Articles in Peer-reviewed Journals	Ly P, Eskiocak U, Kim SB, Roig AI, Hight SK, Lulla DR, Zou YS, Batten K, Wright WE, Shay JW. "Characterization of aneuploid populations with trisomy 7 and 20 derived from diploid human colonic epithelial cells." Neoplasia. 2011 Apr;13(4):348-57. PMID: 21472139 , Apr-2011

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Awards	Eskiocak U, Kim SB, Ly P, Biglione S, Komurov K, Cornelius C, Wright WE, White MA, Shay JW, Roig AI. "Poster of Distinction for 'Stable Down-Regulation of Putative Colorectal Cancer Candidate Genes Identifies Extensive Fragility in the Matrix-Dependent Proliferation Control Network', Digestive Disease Week 2011, Chicago, IL, May 7-10, 2011. http://ddw.scientificposters.com/epsabstract.cfm?id=1 " May-2011